

THYBOROEN, Denmark

WMO: 060520

Lat: 56.7067N

Lon: 8.2150E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.5	-4.0	-8.4	1.8	-3.9	-7.0	2.1	-2.9	18.8	2.7	16.7	5.1	5.5	50	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	24.4	18.4	22.4	17.7	20.9	17.2	19.3	22.9	18.4	21.5	17.6	20.3	5.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	20.9	17.1	12.2	20.1	16.4	11.7	19.3	55.0	23.1	52.0	21.5	49.6	20.4	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.8	13.3	DB	-6.7	27.8	2.1	2.4	-8.3	29.5	-9.5	30.9	-10.7	32.3	-12.3	34.0	(4)
(5)			WB	-7.3	20.6	2.0	1.0	-8.8	21.3	-10.0	21.9	-11.1	22.4	-12.6	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	2.5	2.1	3.7	7.5	11.3	14.2	17.0	17.4	14.8	10.8	6.9	4.1	(6)	
	DBStd	6.05	3.28	3.14	2.65	2.53	2.82	2.38	2.57	2.11	2.03	2.49	2.89	3.50	(7)	
	HDD10.0	1046	232	220	195	84	16	0	0	0	0	21	96	183	(8)	
	HDD18.3	3296	490	453	453	325	221	127	58	44	108	233	342	441	(9)	
	CDD10.0	831	0	0	0	9	54	126	218	230	144	46	4	1	(10)	
	CDD18.3	39	0	0	0	0	1	3	18	16	2	0	0	0	(11)	
	CDH23.3	108	0	0	0	0	3	9	61	33	2	0	0	1	(12)	
	CDH26.7	11	0	0	0	0	0	0	8	2	0	0	0	0	(13)	
(14) Wind	WSAvg	6.7	7.2	6.8	6.6	6.0	6.2	6.5	6.1	6.4	7.0	7.1	7.3	7.3	(14)	
(15) Precipitation	PrecAvg	811	69	43	53	41	49	52	58	76	86	100	100	82	(15)	
	PrecMax	1036	151	123	130	84	232	125	151	175	199	239	203	205	(16)	
	PrecMin	594	9	5	7	0	8	13	8	1	19	24	35	15	(17)	
	PrecStd	121	33	29	30	22	41	30	33	45	46	53	43	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	8.1	11.3	17.6	23.0	24.6	27.8	26.6	22.6	17.3	13.8	10.5	(19)	
		MCWB	7.8	6.7	8.2	11.7	16.0	17.5	19.4	19.7	17.7	14.7	12.7	9.4	(20)	
	2%	DB	8.0	7.2	8.5	14.9	20.0	21.9	25.1	23.9	20.1	15.6	12.3	9.5	(21)	
		MCWB	6.8	6.2	6.7	10.8	14.5	16.2	18.8	18.5	16.7	13.8	11.1	8.4	(22)	
	5%	DB	7.2	6.6	7.4	13.0	17.5	19.4	22.6	21.9	18.5	14.8	11.4	8.9	(23)	
		MCWB	6.2	5.8	6.1	9.9	13.6	15.6	18.1	18.1	15.8	13.2	10.2	7.8	(24)	
	10%	DB	6.6	6.1	6.7	11.2	15.2	17.6	20.8	20.5	17.5	14.0	10.6	8.2	(25)	
		MCWB	5.6	5.2	5.6	8.8	12.3	14.8	17.5	17.3	15.3	12.5	9.3	7.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	7.2	8.3	12.7	17.2	18.4	20.5	20.3	18.4	15.3	12.8	9.6	(27)	
		MCDB	8.7	7.7	10.2	15.9	21.5	22.4	25.5	24.3	21.0	16.7	13.5	10.1	(28)	
	2%	WB	7.1	6.5	7.1	11.2	15.3	17.1	19.4	19.3	17.2	14.2	11.4	8.6	(29)	
		MCDB	7.7	7.0	8.1	14.0	18.8	20.9	23.3	22.9	19.3	15.2	12.1	9.2	(30)	
	5%	WB	6.5	5.9	6.4	10.1	13.8	16.0	18.6	18.4	16.4	13.5	10.4	8.0	(31)	
		MCDB	7.1	6.5	7.2	12.4	16.9	18.9	21.8	21.4	18.2	14.5	11.2	8.7	(32)	
	10%	WB	5.8	5.3	5.7	9.1	12.5	14.9	17.7	17.7	15.7	12.8	9.5	7.4	(33)	
		MCDB	6.5	5.9	6.5	11.0	15.0	17.2	20.5	20.1	17.2	13.9	10.4	8.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	2.8	3.4	4.5	4.6	4.1	4.2	4.1	3.5	3.2	2.8	2.9	(35)	
		MCDBR	2.6	2.5	4.2	7.9	8.7	7.8	8.0	6.6	5.3	3.7	2.7	2.7	(36)	
	5% WB	MCWBR	2.8	2.3	3.0	4.6	4.9	4.2	3.6	3.3	3.0	3.0	2.9	2.9	(37)	
		MCWBR	2.6	2.4	3.7	7.1	7.8	7.0	6.8	5.7	4.8	3.5	2.7	2.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.341	0.358	0.392	0.381	0.376	0.389	0.393	0.373	0.352	0.329	0.304	(40)	
		taud	2.215	2.279	2.324	2.280	2.377	2.416	2.396	2.388	2.427	2.390	2.276	2.201	(41)	
	Ebn at Noon		543	702	796	821	857	866	845	814	776	686	532	457	(42)	
		Edh at Noon	55	77	94	114	110	107	107	102	85	69	52	43	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.42	1.00	2.38	4.02	5.34	5.78	5.33	4.20	2.82	1.40	0.53	0.29	(44)	
	RadStd		0.08	0.16	0.22	0.46	0.51	0.47	0.54	0.32	0.28	0.17	0.09	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	-127	-169	N/A	N/A

Nomenclature: See separate page